

SECTION A (25 marks)

This section consists of 3 questions. Answer **all** questions.

1. Evaluate the following limits:

(a) $\lim_{x \rightarrow -1} \frac{x^3 + 1}{x + 1}$ [3 marks]

(b) $\lim_{x \rightarrow 3} \frac{\sqrt{x+1} - 2}{x - 3}$ [3 marks]

(c) $\lim_{x \rightarrow \infty} \frac{(2-x)(x-1)}{(x-3)^2}$ [3 marks]

2. (a) Find $\frac{dy}{dx}$ if:

(i) $y = \ln[\sin(9 - 2x)]$ [4 marks]

(ii) $y = \tan(e^{2x})$ [3 marks]

(b) Evaluate $\frac{dy}{dx}$ for $y = (x^2 + 1)^5$ when $x = -1$. [4 marks]

3. Given a function $f(x) = 2x^4 - 16x^2 + 14$. By using the second derivative test, find the maximum and the minimum points of $f(x)$. [5 marks]

SECTION B (75 marks)

This section consists of 7 questions. Answer **all** questions.

1. Given $z_1 = 2 + 3i$ and $z_2 = 4 - 4i$. Express $z = \frac{(z_2)}{(z_1)} + \left[\left(\frac{i^3}{-z_2} \right) \right]$ in Cartesian form.

Hence, find the modulus, argument and polar form for z .

[10 marks]

2. Find the solution set of the inequality $\frac{1}{|x-1|} \leq \frac{2}{|x+2|}$.

[10 marks]

3. If given $g(x) = \frac{x+a}{bx+c}$ and $g^{-1}(x) = 1 + \frac{3}{x+1}$, find the values of the constants a , b and c .

[6 marks]

4. The function f and g are defined as

$$f(x) = 3\sqrt{1-x}$$

$$g(x) = \ln\left(\frac{x-2}{3}\right)$$

- (a) If the function $h(x)$ is defined as $h(x) = \begin{cases} f(x) \\ g(x) \end{cases}$ for $x \in R$, sketch $h(x)$ and state the domain and range of $h(x)$.

[4 marks]

- (b) Show that $h(x)$ is not a one to one function.

[2 marks]

- (c) Find $g^{-1}(x)$ and hence, sketch $g^{-1}(x)$ on the same axes with $g(x)$. Hence, state the domain and range of $g^{-1}(x)$.

[6 marks]

5. Consider the function:

$$f(x) = \begin{cases} \frac{x}{x-3} & ; \quad x < -3 \\ x+1 & ; \quad -3 \leq x \leq 3 \\ (kx-3)^2 & ; \quad x > 3 \end{cases}$$

Where $k > 1$.

- (a) Determine whether $f(x)$ is differentiable at $x = -3$. [5 marks]
- (b) Determine the value of k if the function $f(x)$ is continuous at $x = 3$. [4 marks]
- (c) Find the horizontal asymptote of the graph of $f(x)$ as $x \rightarrow -\infty$. [2 marks]

6. (a) Parametric equation of a curve is given by $x = t^3 - 3t$ and $y = 3t^2$.
Find

- (i) $\frac{dy}{dx}$ in terms of t . [4 marks]
- (ii) $\frac{d^2y}{dx^2}$ when $t = \sqrt{3}$. [4 marks]

- (b) Find the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at the point $(1,3)$ on the curve $3x^2 + y^2 = 4y$. [8 marks]

7. Salt water is poured into a right inverted cone with a semi-vertical angle of 45° at a rate of $3\pi \text{ cm}^3 \text{ s}^{-1}$. Find

- (a) The depth of the salt water after 45 seconds. [7 marks]
- (b) The rate of change of its depth at that instant. [3 marks]

END OF QUESTION

Final Answer:

SECTION A

1. (a) 3 (b) $\frac{1}{4}$ (c) -1
2. (a) (i) $-2\cot(9-2x)$ (ii) $2e^{2x}\sec^2(e^{2x})$ (b) -160
3. (0,14) maximum point ; (2,-18), (-2,-18) minimum points

SECTION B

1. $z = \frac{147}{104} + \frac{45}{104}i$; $|z| = \sqrt{\frac{909}{416}}$; $\text{Arg}(z) = 0.2971 \text{ rad}$;
 $z = \sqrt{\frac{909}{416}} [\cos(0.2971 \text{ rad}) + i \sin(0.2971 \text{ rad})]$
2. $(-\infty, 0] \cup [4, \infty)$, $x \neq -2$
3. $a = -4$, $b = -1$, $c = 1$
4. (a) $D_h = (-\infty, 1] \cup (2, \infty)$; $R_h = (-\infty, \infty)$
(b) by using horizontal line test
(c) $g^{-1}(x) = 3e^x + 2$; $D_{g^{-1}} = (-\infty, \infty)$; $R_{g^{-1}} = (2, \infty)$
5. (a) $f(x)$ is not differentiable at $x = -3$.
(b) $k = \frac{5}{3}$
(c) $y = 1$
6. (a) (i) $\frac{dy}{dx} = \frac{2t}{t^2-1}$; $\frac{d^2y}{dx^2} = \frac{-2-2t^2}{3(t^2-1)^3}$; $\frac{d^2y}{dx^2} \Big|_{t=1} = -\frac{1}{3}$
(b) $\frac{dy}{dx} = \frac{3x}{2-y}$; $\frac{dy}{dx} \Big|_{(1,3)} = -3$; $\frac{d^2y}{dx^2} \Big|_{(1,3)} = -12$
7. (a) $h = 7.3986 \text{ cm}$
(b) $\frac{dh}{dt} \Big|_{h=7.3986} = 0.0548 \text{ cm/s}$